



An interpretable Dictionary Learning approach for Diagnostic Hypotheses from 3D Chest

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Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decisionmaking. Current state-of-the-art (SotA) deep learning models often rely on posthoc explainability techniques that provide approximate localizations and lack direct traceability to the models' decisions. Furthermore, SotA models rarely support simultaneous multiclass classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be backprojected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

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Acronyms

3DCT	ThreeDimensional Computed Tomography
AI	Artificial Intelligence
AUROC	Area Under the Receiver Operating Characteristic Curve
CAM	Class Activation Mapping
CNN	Convolutional Neural Network
CT	Computed Tomography
CTRATE	Computed Tomography Report, Annotation, and Text Evaluation
Dice	Dice Similarity Coefficient
DICOM	Digital Imaging and Communications in Medicine
GradCAM	Gradientweighted Class Activation Mapping
HiResCAM	HighResolution Class Activation Mapping
HU	Hounsfield Unit
IoU	Intersection over Union
KSVD	KMeans Singular Value Decomposition
LCKSVD	Label Consistent KSVD
MedMNIST	Medical MNIST
MONAI	Medical Open Network for AI
NIfTI	Neuroimaging Informatics Technology Initiative
ReXGroundingCT	Reportgrounded Explanation and Localization for CT
SotA	StateoftheArt
VLM	VisionLanguage Model
VLM3D	VisionLanguage Modeling for 3D

Chapter 1

Introduction

Developing a unified architectural pipeline for multiclass abnormality classification and grounded localization in 3D Computed Tomography (CT) addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixellevel segmentation as the mechanism for localization forces the architecture to provide voxellevel visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

1.1 Background

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis.

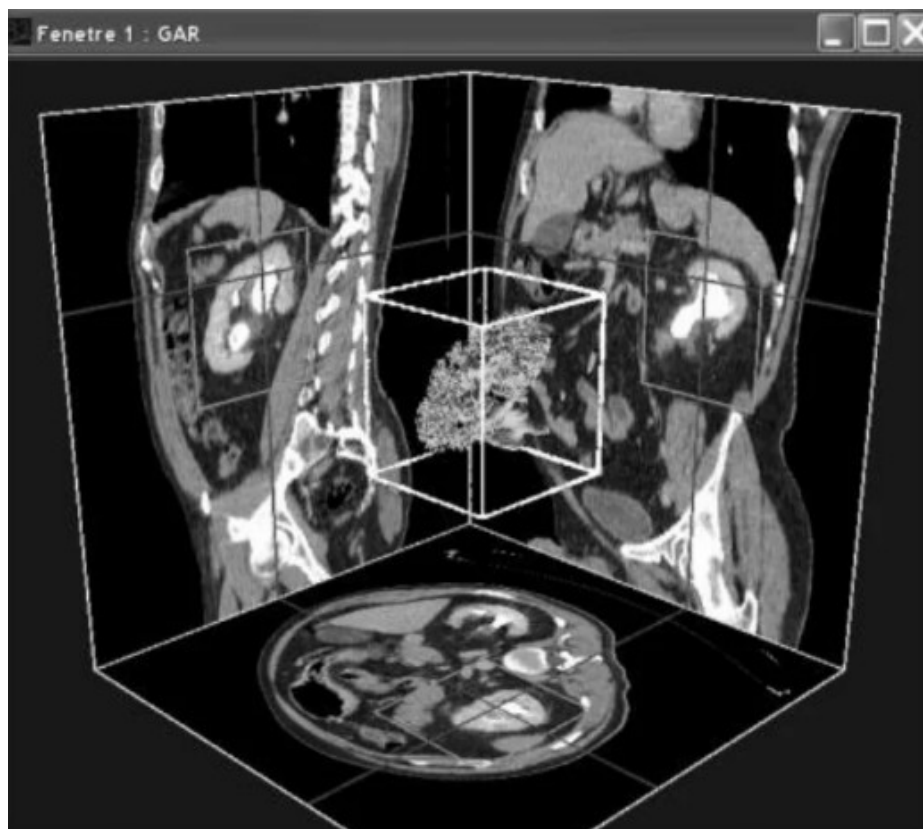


Figure 1.1: Transition from 2D projectional radiography to volumetric CT.

For Lung disease diagnosis, Chest Xrays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormalities is limited when compared to 3D CT. [1].

Furthermore, CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of chest CT volumes has been severely constrained by a lack of such data. However, the recent introduction of large-scale, text-paired datasets has facilitated the development of state-of-the-art

(SotA) vision-language foundation models tailored specifically for volumetric imaging [2–4].

Beyond deep learning approaches, Dictionary Learning algorithms, particularly, Label Consistent KSVD (LCKSVD) [5], Label Embedded Dictionary Learning (LEDL) and Fischer Discriminant Dictionary Learning (FDDL) offer viable approaches for interpretable CT analysis. All three techniques incorporate label information during the dictionary training process, and ensure that the sparse representation of the data is highly discriminative, making them well-suited for multi-abnormality tasks.

1.2 Problem Statement

The clinical utility of Deep Learning-based approaches is hindered by a lack of interpretability. While tools like GradCAM provide posthoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model's decision-making process. In high-stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance.

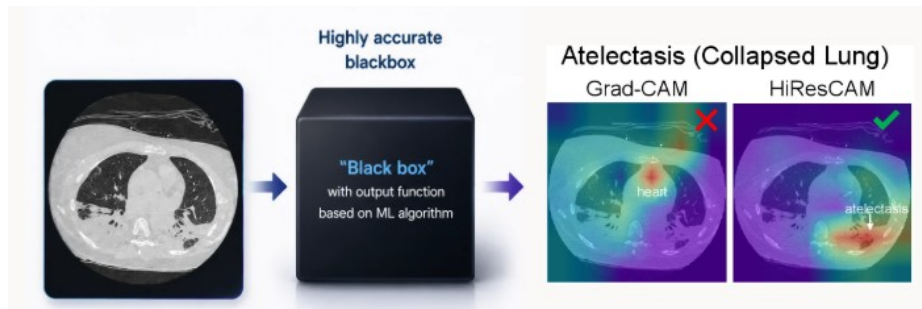


Figure 1.2: HiResCAM example illustrating higher-resolution localization compared with GradCAM.

1.3 Objectives and Scope

The primary goal of this project is to address the limitations of existing Deep Learning-based approaches for Lung abnormality classification and localization by developing an inherently interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning approach to learn normal and pathological dictionary atoms using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.
- Benchmark model performance against stateoftheart classification and segmentation models on the same dataset.
- Evaluate abnormality localization by the model on a realworld dataset with expert radiologist review.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 3 thoracic abnormalities: Pulmonary Nodules, Ground-Glass Opacities, and Consolidations, based on expert radiologist recommendation and dataset availability.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to linear methods, and will not explore nonlinear or hybrid models that provide better performance for less explainability.

1.4 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classificationbased models that prioritize diagnostic labels
- Segmentationbased models that prioritize boundary precision
- Unified architectures designed for grounded localization

SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multitask learning or foundation models that can perform both diagnosis and voxellevel segmentation simultaneously.

Table 1 summarizes the key models in each category, their primary task orientation, and their performance on standard benchmarks.

Table 1.1: Summary of major 3D CT analysis paradigms and representative models

Paradigm	Representative Models	Primary Task	Typical Strengths / Notes
Classificationbased	CTCLIP; Merlin; DINOv3	Imagelevel diagnosis; retrieval; linear probing	Strong imagelevel diagnosis; good zeroshot and transfer performance; efficient inference; limited voxellevel localization.
Segmentationbased	nnUNet; SegVol; SAMMed3D	Voxellevel lesion segmentation	High boundary precision and dense localization; foundation models aim for universal volumetric segmentation across organs and pathologies.
Unified architectures	BiomedParse; MedImageParse; CitrusV; MTMed3D; MedVista3D	Joint diagnosis, segmentation, and reasoning	Multitask inference with grounded localization; integrates report understanding with voxel outputs; typically more complex and computationally intensive.

1.4.1 Classification Architectures

Classificationbased models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D UNet, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and selfsupervised encoders. The benchmark for this paradigm is the CTRATE dataset, which introduced the first public largescale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive LanguageImage Pretraining for Computed Tomography (CTCLIP).

CTCLIP is a VLFM designed for 3D chest CT analysis. It extends the CLIP framework to volumetric medical imaging by jointly learning representations from CT scans and their corresponding radiology reports. [6, 7] Through contrastive learning, the model aligns image features with textual descriptions, enabling tasks such as zeroshot classification, crossmodal retrieval, and reportguided abnormality localization. [6] The model is particularly effective in learning semantic relationships between

radiological findings and volumetric imaging patterns such as GGO, consolidation, fibrosis, and pulmonary nodules. [6, 8]

Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domainspecific pretraining. Merlin is a VLFM exclusively pretrained on abdominal CT scans. Despite it’s training, Merlin has consistently demonstrated exceptional zeroshot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CTCLIP, M3FM, and CTFM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in crossdomain retrieval tasks, Merlin surpassed CTCLIP by 24.7%.

DINOv3, a selfsupervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CTNet and CTCLIP across all metrics on several classification tasks due to it’s ability to capture finegrained textural features.

1.4.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D UNet, such as nnUNet, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAMMed3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAMMed3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA textprompted segmentation models that demonstrate nearperfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and illdefined boundaries of chest CT abnormalities.

1.4.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chainofthought reasoning, enabling pixellevel localization, structured report generation, and physicianlike diagnostic inference in a single framework. BiomedParse, and it’s clouddeployed version MedImageParse, developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include CitrusV, MTMed3D and MedVista3D, where each architecture incorporates a multitask learning (MTL) framework [9–11], as summarized in Table 1.1.

1.4.4 Dictionary Learning Approaches

Applications of sparse dictionary learning in medical imaging have demonstrated significant success across diverse diagnostic tasks. In histopathological analysis, Discriminative Feature oriented Dictionary Learning (DFDL) has shown the ability to learn class specific dictionaries that achieve high classification accuracy even with limited training data, a critical advantage when generous training samples are not available [12]. In neuroimaging, multi feature kernel supervised dictionary learning

approaches have demonstrated superior performance in distinguishing Alzheimer’s disease patients from cognitively unimpaired individuals, achieving 98.18% classification accuracy while maintaining fast testing times [13].

Most relevant to our work, dictionary learning has been successfully applied to lung disease diagnosis from CT images. One approach utilized both frozen dictionary learning and label consistent LCKSVD to classify COVID 19 CT scans, achieving 89% accuracy on the CCCCI dataset [14]. The framework’s key advantage lies in its interpretability, where the sparse coding output can be directly mapped back to the original input signal domain. This work demonstrates that sparse representation methods offer an effective balance between performance and explainability for lung disease classification tasks, providing clear feature representations while maintaining competitive accuracy. Similarly, sparse representation methods have achieved 95.4% accuracy in classifying diffuse lung disease patterns on HRCT images, effectively handling the high variety and complexity of patterns [15].

1.4.5 Explainability Approaches

GradCAM is one of the most widely used posthoc explanation methods for CNNbased classifiers [16]. It generates classdiscriminative heatmaps by using gradients flowing into the final convolutional layer, allowing the user to identify image regions that influenced a prediction [16]. It’s main limitation is that it produces lowresolution saliency maps and does not directly provide precise lesion boundaries or voxellevel localization, which is a major drawback for 3D CT interpretation [16, 17].

SegGradCAM extends this idea by adapting GradCAM to semantic segmentation outputs [18]. Instead of only highlighting regions that are important for imagelevel classification, it produces pixellevel relevance heatmaps for segmentation predictions [18]. This is particularly useful for volumetric medical imaging tasks where the target abnormality occupies a sparse region within a large 3D scan.

HiResCAM addresses one of the key weaknesses of GradCAM by producing higherresolution and more faithful localization maps [17]. This is important in chest CT because many abnormalities, such as small nodules, consolidations, or subtle interstitial changes, may be missed by coarse heatmaps. HiResCAM aims to preserve fine spatial detail, making it more suitable for medical imaging applications where precise localization is required [17]. However, like other gradientbased methods, it remains a posthoc explanation technique and does not change the internal decision process of the model.

1.5 Report Outline

This report is organized into five main chapters.

Chapter 1 introduces the research problem, background, objectives, scope, and literature review related to multiclass abnormality classification and grounded localization in 3D chest CT imaging. It also discusses existing deep learning and dictionary learning approaches together with their limitations in explainability.

Chapter 2 presents the theoretical background required for this study, including sparse representation, dictionary learning, KSVD, LCKSVD, frozen dictionary learning, and Vision Transformers used in medical imaging.

Chapter 3 describes the proposed methodology of the project. This includes dataset acquisition, preprocessing steps, patch extraction, implementation of the LCKSVD framework, abnormality classification pipeline, and the proposed backprojection mechanism used for interpretable localization and explainability generation.

Chapter 4 explains the experimentation and evaluation framework used in this research. It presents the selected baseline models, evaluation metrics, and the three evaluation axes: classification accuracy, grounded localization quality, and explainability. The chapter also summarizes the current progress of the project and planned evaluations.

Finally, Chapter 5 discusses the overall findings, contributions, limitations, and future improvements of the proposed interpretable dictionary learning framework for 3D chest CT abnormality analysis.

Chapter 2

Background

2.1 Dictionary Learning

2.1.1 Sparse Coding and KSVD

Sparse coding is the process of representing an input signal as a linear combination of only a small number of basis elements selected from a dictionary. In the context of this project, each input signal corresponds to a vectorised 3D CT patch. For example, a patch of size $32 \times 32 \times 32$ contains 32,768 voxel values and is reshaped into a column vector before being used by the dictionary learning model.

Given an input patch $y \in \mathbb{R}^m$ and a dictionary $D \in \mathbb{R}^{m \times K}$ containing K atoms, sparse coding aims to find a coefficient vector $x \in \mathbb{R}^K$ such that:

$$y \approx Dx$$

where D contains learned dictionary atoms as its columns, and x is the sparse code that indicates how strongly each atom contributes to reconstructing the input patch. The term “sparse” means that most values in x are zero, and only a small number of atoms are active for a given patch.

The sparse coding problem can be formulated as:

$$x = \arg \min_x \|y - Dx\|_2^2 \quad \text{s.t.} \quad \|x\|_0 \leq T$$

where $\|y - Dx\|_2^2$ measures the reconstruction error, $\|x\|_0$ counts the number of non-zero coefficients in the sparse code, and T controls the maximum number of atoms that can be used to represent one patch. A smaller value of T forces the model to use fewer atoms, resulting in a more compact and interpretable representation.

The dictionary D is typically learned using K-SVD (K-Singular Value Decomposition), which solves the following optimization problem:

$$\min_{D, X} \|PDx\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T$$

where P represents the training data, X is the sparse coefficient matrix, and T controls the sparsity level. KSVD alternates between sparse coding and dictionary update steps to minimize reconstruction error. However, it does not incorporate class labels, which limits its discriminative ability for classification tasks. [19]

2.1.2 LCKSVD

LCKSVD jointly learns a dictionary, sparse codes, a labelconsistency transform, and a linear classifier so that the learned representations are not only good for reconstruction but also discriminative for classification through labelconsistency constraints [5].

The standard LCKSVD objective (often referred to as LCKSVD2 when the classifier term is included) can be written compactly as:

$$\min_{D,A,W,X} \|PDX\|_F^2 + \alpha \|QAX\|_F^2 + \beta \|HWX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T, \|d_k\|_2 = 1 \quad \forall k \quad (2.1)$$

Definitions of terms:

- $P \in \mathbb{R}^{m \times N}$: training data matrix (each column is a vectorised 3D patch).
- $D \in \mathbb{R}^{m \times K}$: dictionary matrix with K atoms (columns d_k).
- $X \in \mathbb{R}^{K \times N}$: sparse coefficient matrix whose i th column x_i encodes the i th patch.
- T : sparsity threshold (maximum nonzeros allowed per sparse code).
- $Q \in \mathbb{R}^{p \times N}$: labelconsistency target matrix that embeds class information into a prototype space used to regularise codes.
- $A \in \mathbb{R}^{p \times K}$: linear transform that maps sparse codes to the labelconsistency space (enforces that codes for the same class are similar).
- $H \in \mathbb{R}^{c \times N}$: label indicator matrix (onehot encoding across c classes) used for classifier training.
- $W \in \mathbb{R}^{c \times K}$: linear classifier weights mapping sparse codes to class scores.
- α, β : nonnegative scalars controlling the strength of the labelconsistency and classification terms respectively.
- $\|\cdot\|_F$: Frobenius norm; $\|x\|_0$ denotes the ℓ_0 (sparsity) pseudonorm.

Role of each objective term:

- $\|PDX\|_F^2$: reconstruction fidelity; forces the dictionary and codes to accurately represent the input patches.
- $\alpha \|QAX\|_F^2$: labelconsistency regulariser; encourages codes for the same class to be mapped close to a shared prototype.
- $\beta \|HWX\|_F^2$: supervised classification loss on sparse codes; trains a linear classifier jointly with the dictionary.

In practice LCKSVD is solved by alternating optimisation: (1) sparse coding (e.g., OMP) to estimate X with D, A, W fixed, and (2) closedform or SVDbased updates for D, A, W while holding X fixed. Variants differ in how Q is constructed and whether the classifier term is included during early warmstart iterations.

2.1.3 Frozen Dictionary Learning

To address this limitation, we adopt a twostage dictionary learning strategy inspired by constrained dictionary learning approaches. [20] In the first stage, a dictionary is learned from normal CT scans to capture healthy anatomical structures. In the second stage, this dictionary is frozen and kept fixed, while additional atoms are learned only from abnormal scans. This approach can be interpreted as:

- preserving D_{normal} learned from healthy data
- learning D_{abnormal} independently for disease patterns

This separation ensures that normal anatomical structures are not overwritten, while abnormal features are explicitly isolated, improving interpretability and stability.

2.2 Radiological Patterns in Lung CT Imaging

2.2.1 CT Imaging

Computed Tomography (CT) imaging is widely used in lung disease diagnosis due to its ability to generate detailed crosssectional images of lung structures. [21] CT scans provide highresolution visualization of abnormalities such as Ground Glass Opacity (GGO), consolidation, Atelectasis, and pulmonary nodules, enabling accurate assessment and early detection of lung diseases. [22]

In modern medical imaging systems, CT imaging also supports computeraided diagnosis and deep learningbased analysis for automated detection, segmentation, and classification of lung abnormalities. [23] These capabilities improve diagnostic efficiency and assist radiologists in clinical decisionmaking.

2.2.2 Ground Glass Opacity (GGO)

GGO appears as hazy regions in CT images which partially obscure underlying blood vessels or airway structures. Typically indicates partial filling of air spaces, inflammation, or earlystage infection. [24]

2.2.3 Consolidation

Consolidation appears denser than GGO, completely obscuring underlying lung structures such as vessels and airway walls. [22] It is often associated with severe infection, fluid accumulation, or advanced disease stages. The transition between GGO and consolidation is clinically important, as it can indicate disease progression or worsening lung conditions. [25]

2.2.4 Atelectasis

Atelectasis refers to the collapse or closure of alveoli in the lungs, leading to reduced lung volume and impaired gas exchange. It can be caused by various factors, including mucus plugging, foreign body aspiration, or compression from external masses. On CT imaging, atelectasis appears as areas of increased density with volume loss and may be associated with air bronchograms. [22]

2.2.5 Pulmonary Nodules

Pulmonary nodules are small, rounded lesions commonly observed in lung CT scans and are important indicators for early disease detection. They're usually less than 3 cm in diameter and can appear solid, subsolid, or groundglass in nature. [26] Nodules are clinically significant as they may represent benign conditions such as infections or scars, but can also indicate early stage lung cancer in suspicious cases. [14, 15] Careful monitoring of nodule size, shape, and growth over time is essential for accurate diagnosis.

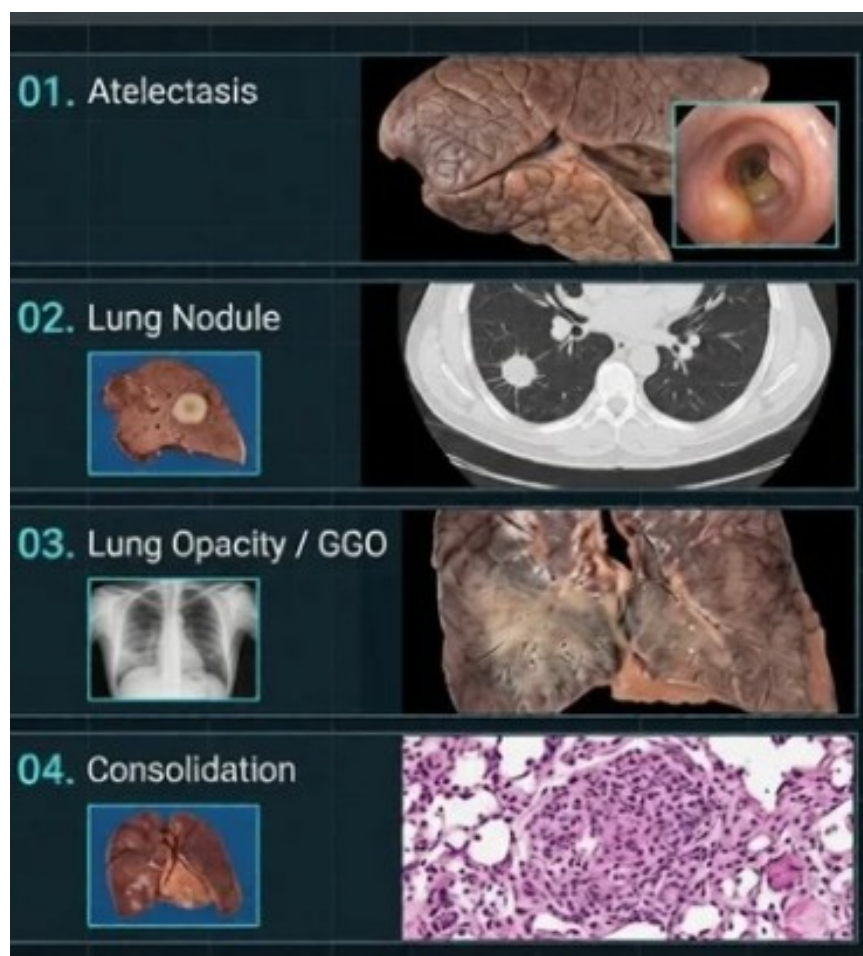


Figure 2.1: Four pathologies: example appearances of Lung nodules, groundglass opacity (GGO), consolidation, and Atelectasis.

Chapter 3

Methodology

We implement a sixstage pipeline for our study, consisting of:

1. Data Acquisition
2. Data Preprocessing
3. Implementation of the LCKSVD Approach
4. Abnormality Classification
5. BackProjecting ClassDiscriminative Atoms and Deriving Explainability through Segmentation Masks
6. Evaluation and Benchmarking

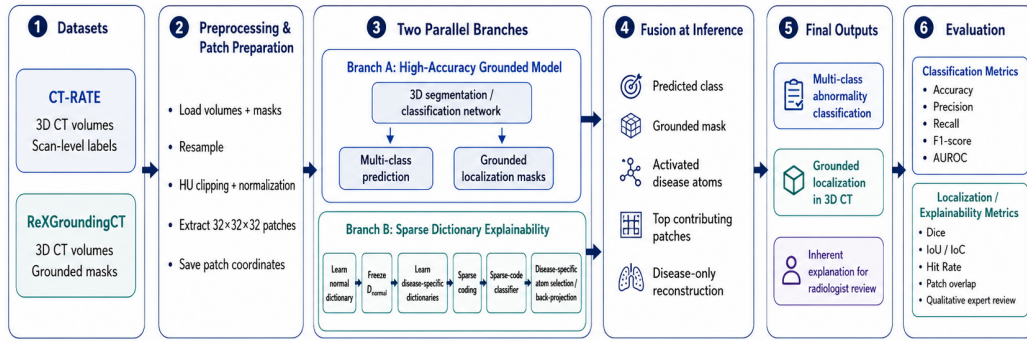


Figure 3.1: Overall methodology pipeline used in the study.

3.1 Data Acquisition

Two public chest CT datasets: CTRATE[27] and ReXGroundingCT[8] were used in our work. CTRATE was used as the primary source of CT volumes (normal and abnormal). while ReXGroundingCT was used to obtain voxellevel segmentation masks per abnormality scan together abnormality labels, for each corresponding CT volume. Although CTRATE also contains abnormality labels, RexGroundingCT was chosen for its more clinically relevant set of findings, as well as its detailed voxellevel annotations that are essential for our explainability evaluation.

After an initial review of the thoracic findings available in ReXGroundingCT, a discussion with the supervising radiologist produced a shortlist of the following

abnormalities that are clinically significant and have sufficient representation in the datasets:

1. Atelectasis
2. Lung nodule
3. GroundGlass Opacity (GGO)
4. Consolidation

Both the volumes and their segmentation masks are formatted as NIfTI files. The masks are 3D volumetric segmentations ($H \times W \times D$) that align with the corresponding 3D CT scan, with their shape generally aligned with the scan’s dimensions, as [512, 512, Depth]. For scans with multiple findings, masks can be structured as [F, H, W, D], where F represents the number of findings. Within a mask file, pixels are labeled with integer values corresponding to different entities for that finding, while 0 represents background.

3.2 Data Preprocessing

The data preprocessing pipeline consists of four major steps:

1. Resampling to isotropic spacing
2. Intensity windowing
3. Normalization
4. Patch extraction and vectorization

Each volume is resampled to 1.5 mm isotropic using trilinear interpolation to standardize the spatial resolution across scans. If the current spacing is already within 1% of target, this step is skipped. Each mask is also resampled to exactly match the resampled volume’s spatial shape using nearestneighbour interpolation to preserve integer label values

In the second step, a standard window Hounsfield unit (HU) of [1000, +200] was used to separate the lung window from the CT volume. Then, a linear rescaling to [0, 1] as float32 was applied to the windowed volume, to normalize the intensity values.

Finally, the preprocessed volumes are divided into overlapping 3D patches of size $32 \times 32 \times 32$ with 32,768 features per patch (48 mm^3 at 1.5 mm spacing). Normal and abnormal scans are processed in separate phases to ensure that the Dictionary Learning model first learns a representation of normal anatomy before being exposed to abnormalities.

In phase 1, patches are extracted on a regular non-overlapping grid with stride of 32 voxels from each normal scan. Only patches with at least 90% lung mask coverage (`MIN_LUNG_COVERAGE = 0.90`) are retained. All patches are labelled with class index 0 (normal).

In phase 2, using the abnormal scans, we build a percategory binary mask by ORing all finding slots of the same category together; build a union mask across all categories; sample patch centres uniformly from the union mask foreground; each

candidate patch must have $\geq 5\%$ overlap with at least one category mask; the label is assigned to whichever category mask has the highest overlap with that patch (majorityclass rule), which handles overlapping findings correctly

Both phases share the same maximum patches/scan budget; class balance is therefore determined by the normal/abnormal scan counts in the dataset, not a hard ratio. All patches are assembled into X ($4096 \times N$, float64) and H (N ,) int64, then saved as .npz files.

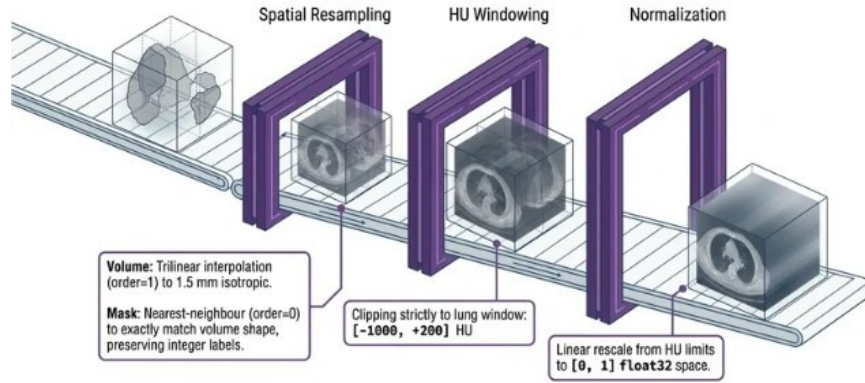


Figure 3.2: Preprocessing pipeline overview.

3.3 Implementation of the LCKSVD Approach

To implement the Dictionary Learning approach for our study, we required a concrete implementation of the LCKSVD algorithm. [5] Currently, many of the available standard implementations of KSVD and LCKSVD are developed using MATLAB, which is not ideal for integration with our Pythonbased data processing and evaluation pipeline. Although A few opensource Python implementations LCKSVD are available, they either lacked an importable library, or did not provide the flexibility required for our study.

Therefore, we developed a custom implementation of the LCKSVD algorithm in Python and published it on PyPI, the official thirdparty software repository for Python, under the name “reppi.” The implementation follows a modular design, consisting of four separate components:

1. SRC Utility Functions shared across Sparse Coding Algorithms
2. OMP Implementation of BatchOMP and OMPCholesky approaches for Sparse Coding
3. KSVD Implementation of KSVD Dictionary Learning Algorithm
4. LCKSVD Implementation of LCKSVD1 and LCKSVD2 Dictionary Learning Algorithms

This design also allows for flexibility in integrating alternative sparse coding methods in the future. The source code is available at github.com/ckekula/reppi.

This modular architecture enables straightforward integration of the algorithm through library imports, while providing the flexibility required to experiment with

different hyperparameters and configurations. The implementations of the SRC, OMP, and LCKSVD modules are provided in Appendix A, Appendix B, and Appendix C, respectively.

3.4 Model Training and Inference

Prior to training, each patch vector is L2normalised to unit norm, ensuring that the sparse coding step operates on texture direction rather than absolute intensity magnitude. Patches with a norm below 1×10^{10} , corresponding to uniform zero regions with no signal, are discarded entirely.

The model is trained using LCKSVD2, a dictionary learning algorithm that jointly optimises three objectives: reconstruction accuracy, label consistency, and linear classification. Training proceeds in two stages: 20iteration KSVD warmstart that initialises the dictionary on reconstruction error alone, followed by 50 LCKSVD2 iterations that refine it under the combined objective. This twostage procedure avoids poor local minima that arise when the labelconsistency and classifier terms are applied before the dictionary has converged to a geometrically meaningful initialisation.

The key hyperparameters are a dictionary size of $K = 128$ atoms, a sparsity constraint of $T = 10$ nonzero coefficients per patch (approximately 8% of K), a label-consistency weight of $\alpha = 4.0$, and a classifier weight of $\beta = 2.0$. On smaller datasets where the patch count N is insufficient to support 128 atoms, the dictionary size is adaptively capped at $\max(8, N/2)$ and the sparsity constraint is scaled proportionally.

At inference time, a new CT volume is loaded and preprocessed through the identical pipeline used during training. However, unlike training, instead of randomly sampling patches as in training, inference requires spatial exhaustiveness so that every region of the volume receives a classification score.

This is achieved by extracting patches on a dense regular grid with a stride of half the patch width along each spatial axis. For a volume of size 512^3 , this produces $32^3 = 32768$ patches, each a 32^3 voxel window centred on a grid point. The 50% overlap between adjacent patches ensures that every voxel is covered by up to 8 patch windows, providing multiple independent score estimates per spatial location for the subsequent backprojection step.

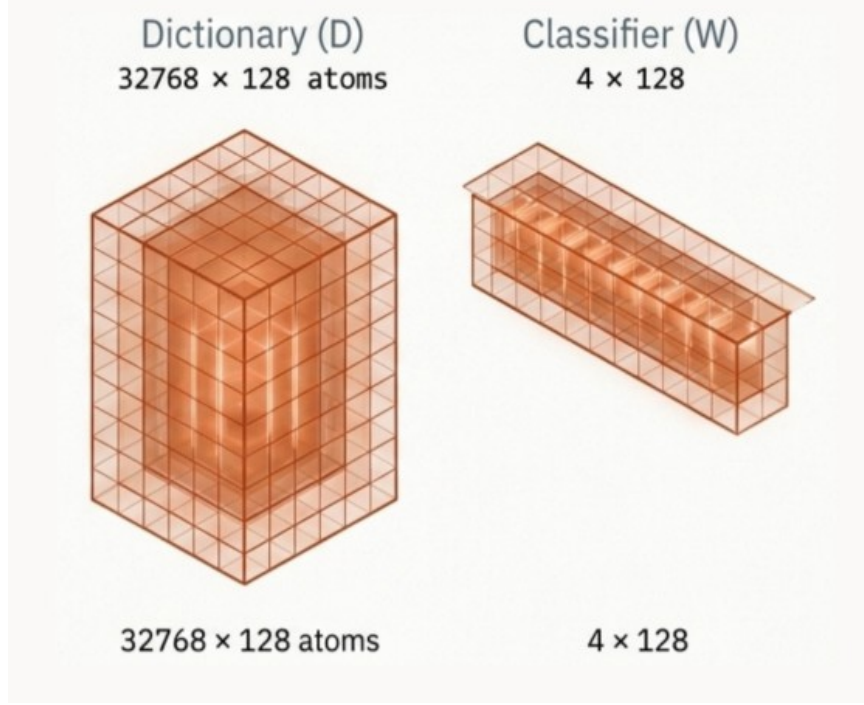


Figure 3.3: Patch extraction grid used during training and dense inference sampling.

3.5 BackProjecting ClassDiscriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through backprojection. Because each atom in the dictionary D is associated with a specific class label through the matrix Q [5], the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class c , the algorithm identifies all atoms d_k that have been assigned to that class. For a query voxel or patch with sparse coefficients x , the contribution C_c is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process backprojects the class-specific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. For classdiscriminative atoms operating on raw voxelspace inputs, the backprojection is an exact, closedform spatial attribution.

3.6 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LCKSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

GradCAM’s approximation has two sources of imprecision — the gradient itself is a local linear approximation of a nonlinear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LCKSVD’s contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive GradCAM approach, atom backprojection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There’s no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate posthoc analysis run afterwards.

Chapter 4

Evaluation Framework and Results

4.1 Evaluation Framework Design

Comparing a sparse linear approach with stateoftheart nonlinear model architectures using only raw performance scores can give an incomplete view of model behavior. In medical imaging, a useful model should not only predict the correct abnormality, but also identify the relevant anatomical region and provide a meaningful explanation for its decision. Therefore, the evaluation framework in this study is organized around three complementary evaluation axes:

- **Axis A: Classification Accuracy** evaluates the ability of the model to detect the presence or absence of each abnormality using standard discriminative metrics such as AUROC, precision, recall, and F1 score.
- **Axis B: Grounded Localization Quality** evaluates whether the model localizes the abnormality in the correct region of the 3D CT volume using spatial overlap and hitbased localization metrics such as DSC, IoU, and HIT@K.
- **Axis C: Interpretability and Explanation Faithfulness** evaluates whether the model’s explanation or attribution map is spatially aligned with the true abnormality region using metrics such as AttributionMask IoU, Pointing Game Accuracy, and Energy Inside Mask.

Axis C is particularly important for this study because the proposed LCKSVD-based approach is designed to provide interpretable decisionmaking through dictionary atoms and sparse activations. In contrast, most deep learning and foundation models require posthoc attribution techniques such as GradCAM, HiResCAM, or saliencybased methods to explain their predictions. These posthoc methods provide useful approximations, but they do not directly represent the internal decision process of the model. Therefore, the evaluation framework explicitly separates predictive performance, localization quality, and explanation faithfulness to better highlight the strengths and limitations of each model type.

4.2 Evaluation Metrics

A common set of quantitative metrics was used to evaluate the models across the three axes. This ensures that different models are compared under the same evaluation

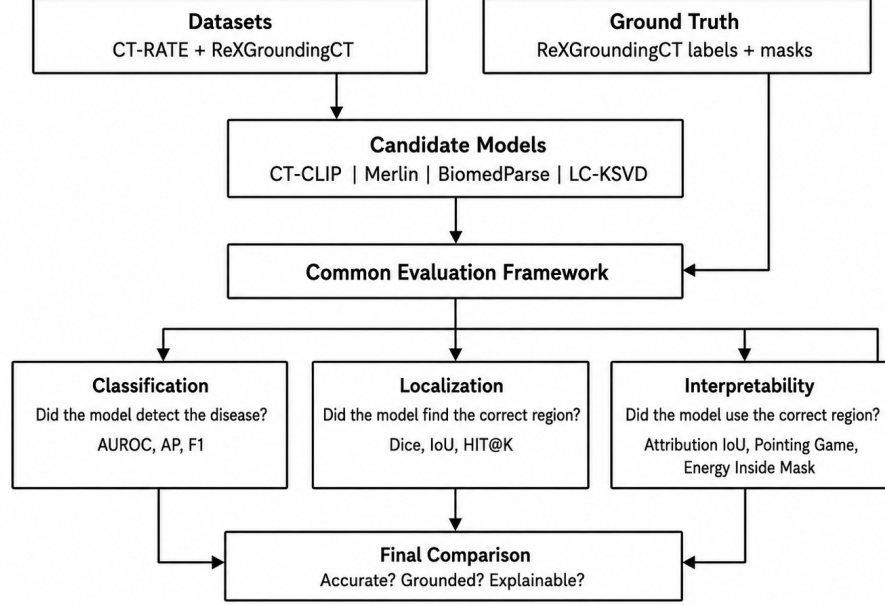


Figure 4.1: Architecture diagram of the proposed evaluation framework.

criteria, even though their raw outputs may differ. Classification-oriented models are evaluated using disease probability scores, localization-oriented models are evaluated using predicted masks, and interpretable models are additionally evaluated using attribution or activation maps.

4.2.1 Classification Metrics

For Axis A, the classification task measures whether the model correctly detects each abnormality. The following metrics are used:

- **Area Under the Receiver Operating Characteristic Curve (AUROC):** AUROC measures how well the model separates positive and negative cases across different decision thresholds. It is useful for imbalanced medical datasets because it evaluates ranking performance rather than relying on a single fixed threshold.
- **Precision and Recall:** Precision measures how many predicted positive cases are actually positive, while recall measures how many true positive cases are correctly detected by the model.
- **F1 Score:** F1 score is the harmonic mean of precision and recall. It provides a balanced measure when both false positives and false negatives are important.
- **Average Precision:** Average precision summarizes the precision-recall curve and is useful when positive cases are relatively rare.

4.2.2 Localization and Grounding Metrics

For Axis B, the localization task measures whether the predicted abnormality region overlaps with the groundtruth lesion mask. This is important because a model may

correctly predict the presence of a disease while still focusing on the wrong anatomical area.

- **Dice Similarity Coefficient (DSC):** Dice measures the spatial overlap between the predicted mask and the groundtruth mask. It is defined as:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

where X is the predicted mask and Y is the groundtruth mask.

- **Intersection over Union (IoU):** IoU, also known as the Jaccard index, measures the ratio between the overlapping region and the total combined region:

$$IoU = \frac{|X \cap Y|}{|X \cup Y|}$$

- **HIT@K:** HIT@K measures the proportion of cases where the model achieves at least a small amount of meaningful overlap with the groundtruth lesion region. In this work, HIT@5, HIT@10, and HIT@25 are used to indicate whether the Dice score exceeds 0.05, 0.10, and 0.25 respectively. This is useful for diffuse abnormalities where exact boundary agreement is difficult.

4.2.3 Interpretability Metrics

For Axis C, the interpretability task evaluates whether the model’s explanation is focused on the clinically relevant abnormality region. This is different from localization because the goal is not only to check where the model predicts a lesion, but also to determine whether the model’s decision is supported by the correct anatomical evidence.

- **AttributionMask IoU:** This measures the overlap between the thresholded attribution map and the groundtruth lesion mask. A higher value indicates that the explanation is spatially aligned with the true abnormality.
- **Pointing Game Accuracy:** This checks whether the highest attribution point lies inside the groundtruth lesion region. If the maximum response is inside the lesion mask, the case is counted as a hit.
- **Energy Inside Mask:** This measures the percentage of total attribution energy that lies inside the groundtruth mask. A high value indicates that the model’s explanation is concentrated on the true abnormality region rather than irrelevant background structures.

4.3 Classification Accuracy Evaluation

The classification performance was evaluated across four abnormality classes: Lung Nodule, Lung Opacity, Consolidation, and Atelectasis. The proposed LCKSVD-based baseline was compared with three foundation model baselines: Merlin, CTCLIP, and BiomedParse. For LCKSVD, consolidation-related performance was also used as a proxy reference for atelectasis due to current dataset and label constraints.

4.3.1 Overall Classification Performance

Figure 4.2 shows the macrolevel classification performance across the evaluated models. Merlin achieved the strongest discriminative performance, with a macroAUROC of 0.966. In contrast, CTCLIP and BiomedParse produced lower macroAUROC values of 0.538 and 0.509 respectively on the ReXGroundingCT evaluation setting.

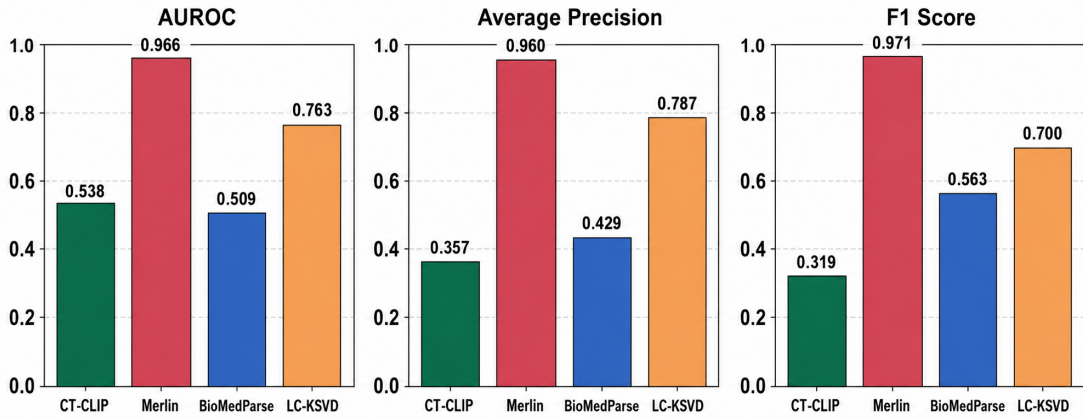


Figure 4.2: Overall classification performance comparison across models.

These results suggest that Merlin is the strongest classification baseline in this setup. CTCLIP and BiomedParse show weaker classification performance, which may be due to differences in training objectives, dataset alignment, and the use of zeroshot or proxy classification outputs. BiomedParse is mainly designed for textguided segmentation and grounding, so its classification score should be interpreted as a proxy rather than as the output of a dedicated classifier.

4.3.2 PerClass Analysis

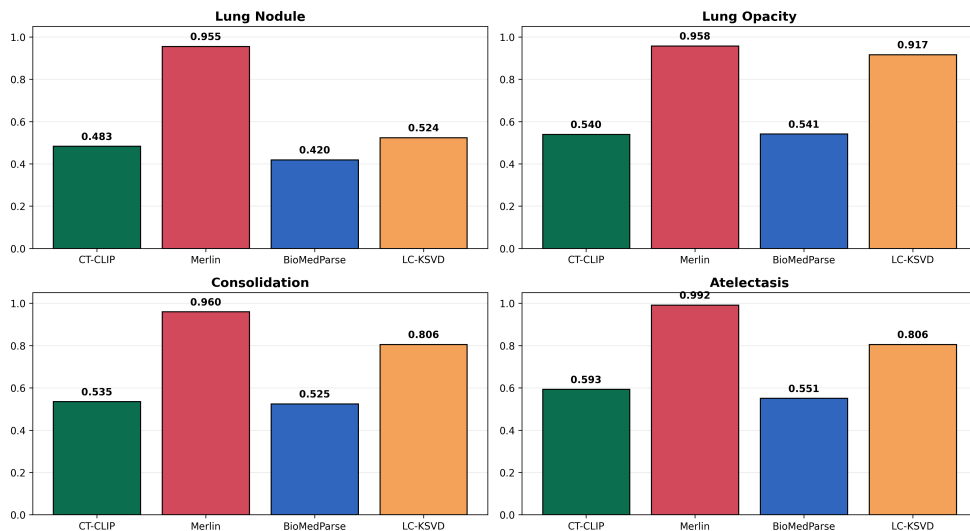


Figure 4.3: Perclass AUROC comparison Across Models.

The perclass AUROC comparison in Figure 4.3 provides a more detailed view of model behavior. Merlin maintains strong performance across all evaluated classes.

The LCKSVD baseline shows promising results for nonfocal or broader abnormalities, achieving AUROC values of 0.917 for Lung Opacity and 0.806 for Consolidation. It also achieves 0.806 for Atelectasis in the current evaluation setting. However, its performance for Lung Nodule is weaker, with an AUROC of 0.524.

This behavior suggests that LCKSVD may capture broader abnormal tissue patterns more effectively than small focal lesions. Lung nodules are compact and localized, so their detection may require finer spatial resolution, multiscale patch modeling, or more nodule-specific training samples. In contrast, abnormalities such as lung opacity and consolidation may produce more distributed visual patterns that are more compatible with the sparse dictionary representation.

4.4 Grounded Localization Evaluation

Axis B evaluates whether the models can identify the correct abnormal region in the volumetric CT scan. This is different from classification because a model can correctly detect the presence of a disease while still localizing the wrong anatomical region. The localization evaluation therefore uses Dice, IoU, and HIT@K metrics to measure spatial agreement between predicted lesion regions and groundtruth annotations.

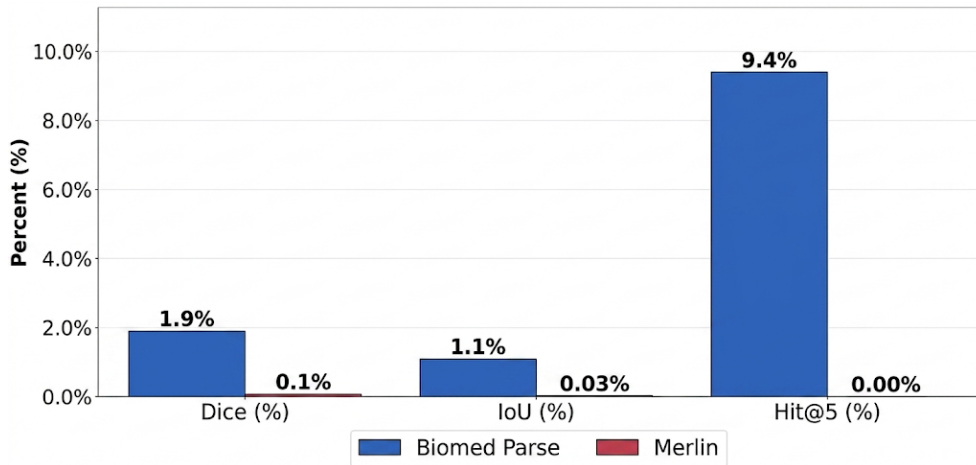


Figure 4.4: Overall localization performance comparison for BiomedParse and Merlin.

Figure 4.4 shows that grounded localization remains challenging for the evaluated models. The Dice, IoU, and HIT@K values indicate limited agreement between predicted lesion regions and reference annotations. This highlights an important gap between disease recognition and true spatial grounding. For clinical support, it is not sufficient to detect an abnormality; the model should also identify the correct location and extent of the abnormal region.

4.4.1 PerClass Localization Performance

Figure 4.5 shows the perclass Dice comparison. The results indicate that localization performance varies substantially across lesion categories. This variation is likely influenced by lesion morphology, size, contrast, and boundary clarity.

Focal lesions such as lung nodules show comparatively better localization than diffuse abnormalities such as lung opacity, consolidation, and atelectasis. This suggests

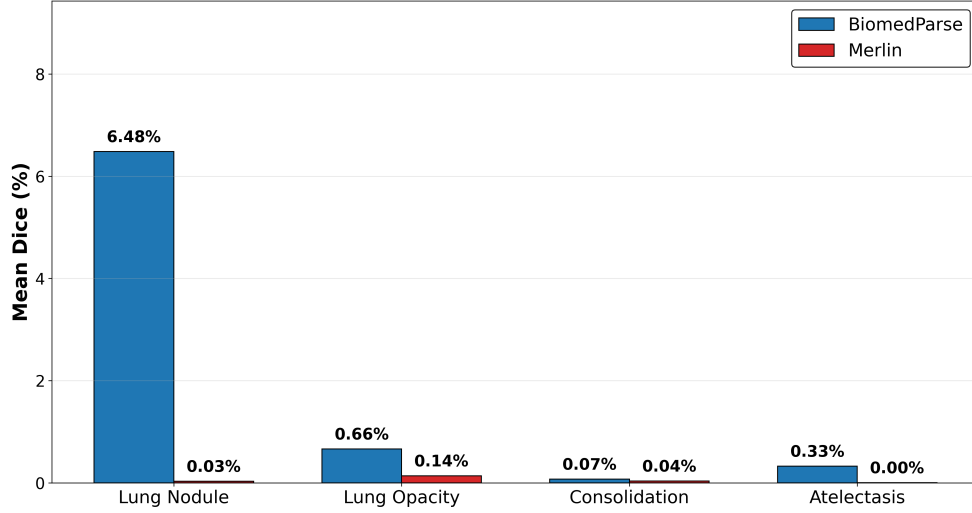


Figure 4.5: Mean Dice comparison by lesion class.

that objectlike lesions with clearer boundaries are easier for textguided segmentation models to localize. In contrast, diffuse abnormalities often have ambiguous borders and variable appearance, making exact spatial overlap more difficult.

4.4.2 Qualitative Comparison of Localization Outcomes

The qualitative BiomedParse example in Figure 4.6 further illustrates the gap between abnormality detection and precise grounding. In the consolidation example, the predicted region covers a broad area of the lung, while the groundtruth annotation is more localized. This indicates that the model responds to the general lung region associated with the abnormality, but it does not accurately preserve the lesion boundary, extent, or class-specific spatial location.

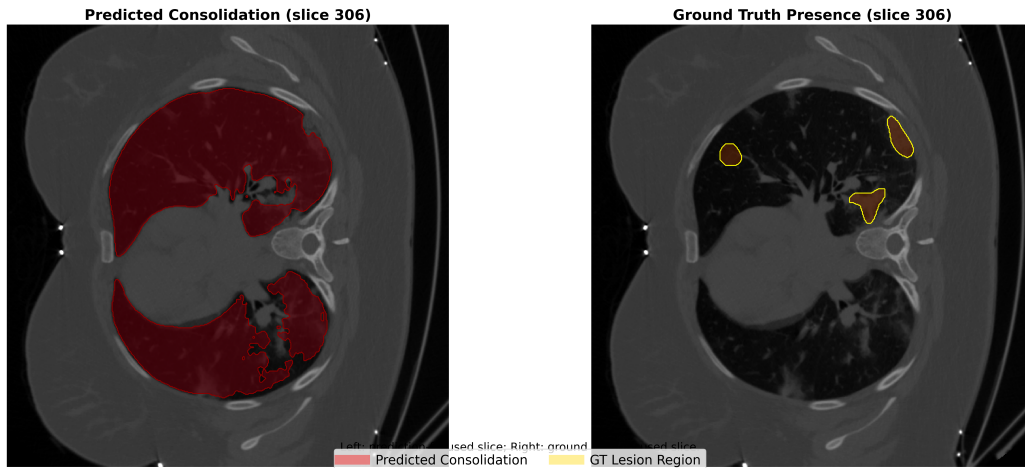


Figure 4.6: Groundtruth and BiomedParsepredicted localization for consolidation.

The LCKSVD visualization in Figure 4.7 shows a related limitation. The prediction map highlights a suspicious region in the lung, indicating some sensitivity to abnormal tissue patterns. However, the activated region is not fully aligned with the groundtruth lesion location. This suggests that the sparse representation cap-

tures some abnormal visual structure, but further refinement is needed to improve anatomical alignment.

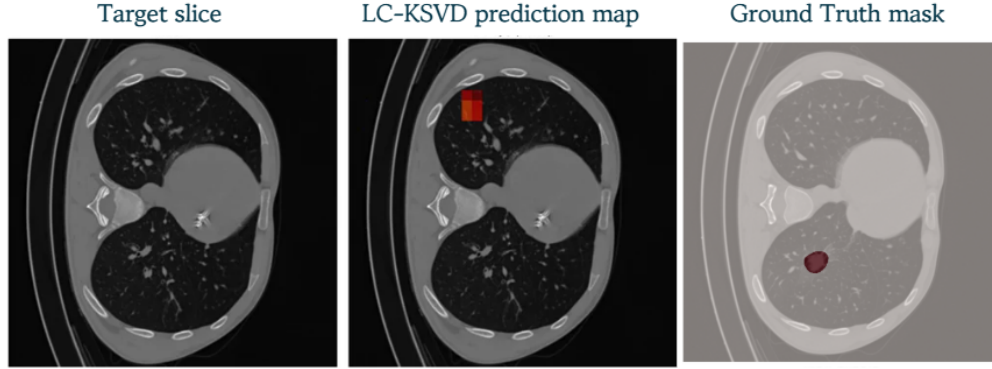


Figure 4.7: Groundtruth and LCKSVDpredicted localization for consolidation.

Overall, the localization results show that both foundation models and the proposed sparse representation approach face challenges in producing precise lesionlevel grounding in 3D CT. This supports the need for evaluation beyond classification scores alone.

4.5 Inherent Explainability Evaluation

Axis C focuses on whether model predictions are supported by clinically meaningful evidence. This is a key motivation for the proposed LCKSVD approach. Deep learning models such as Merlin, CTCLIP, and BiomedParse can be evaluated with posthoc explanation methods or grounding maps, but these explanations are not always part of the model’s internal decision process. LCKSVD, in contrast, provides a more direct form of traceability through its dictionary atoms and sparse coefficients.

4.5.1 AtomtoVoxel Traceability

LCKSVD represents input patches using a learned dictionary and sparse activation coefficients. Since the reconstruction and classification processes are based on linear combinations of dictionary atoms, the contribution of selected atoms can be traced back to the corresponding voxel regions in the CT volume. This provides a direct connection between the model’s prediction and the spatial image patterns that contributed to it.

This atomtovoxel traceability is useful because it allows the model’s decision to be inspected in terms of activated dictionary components. Instead of relying only on a final disease score, the user can examine which atoms were activated and where their contributions appear in the volume.

4.5.2 Spatial Faithfulness of Explanations

The explanation quality is evaluated using attribution maps generated from model activations or atom contributions. These maps are compared with the groundtruth lesion masks using the following metrics:

- **AttributionMask IoU:** Measures how much the explanation map overlaps with the true lesion region.
- **Pointing Game Accuracy:** Checks whether the strongest explanation point falls inside the annotated lesion region.
- **Energy Inside Mask:** Measures how much of the explanation signal is concentrated inside the groundtruth mask.

Together, these metrics assess whether the explanation is spatially faithful to the disease region. A model with good interpretability should not only predict the correct abnormality, but should also base its decision on the relevant anatomical evidence.

For LCKSVD, these metrics are particularly meaningful because the explanation comes from the model structure itself through sparse codes and dictionary atoms. Even when its classification performance is lower than that of a large foundation model, its explanations can provide additional value by making the decision process more transparent and easier to inspect.

4.6 Summary of Evaluation Findings

The evaluation framework provides a multidimensional comparison of model performance. Axis A shows that Merlin is the strongest classification baseline, while LCKSVD demonstrates promising performance for lung opacity, consolidation, and atelectasis. Axis B shows that localization remains difficult for all evaluated models, especially for diffuse abnormalities. BiomedParse performs relatively better on focal lung nodules, but its localization performance is inconsistent across classes. Axis C highlights the main advantage of LCKSVD: its ability to provide interpretable atombased explanations that can be traced back to voxellevel regions.

Overall, the results indicate that classification accuracy alone is not sufficient for evaluating medical AI models in 3D chest CT analysis. A reliable model should be accurate, spatially grounded, and explainable. The proposed evaluation framework captures these three requirements and provides a fair basis for comparing sparse interpretable models with modern foundation model baselines.

Chapter 5

Future Work

There are several improvements that can be made in the future to further develop this research and improve the overall performance of the proposed system.

5.1 Further Improvement of the LC-KSVD Model

More experiments can be carried out to improve the performance of the LC-KSVD model by changing different training parameters such as dictionary size, sparsity constraints, and regularization values. Testing different parameter combinations may help achieve better classification and localization results.

5.2 Studying Other Dictionary Learning Techniques

This study mainly focused on the LC-KSVD approach. In future research, other dictionary learning methods such as K-SVD, D-KSVD, and FDDL can also be tested and compared. This would help identify which method works best for multi-abnormality detection in 3D CT images.

5.3 Validation with Radiologists

Another important future step is to evaluate the model outputs with the support of radiologists. Their medical expertise can help verify whether the detected abnormal regions and predictions are clinically meaningful and accurate. Such evaluations can improve the practical value of the proposed system.

5.4 Comparison with Recent Advanced Models

Future studies can also compare the proposed approach with newer state-of-the-art models developed for medical image classification and localization. This comparison can provide a clearer understanding of the advantages and limitations of the LC-KSVD-based framework.

5.5 Improving Explainability and Reliability

Future work can focus on making the system more explainable and reliable for real clinical use. Providing clearer visual evidence and more stable predictions may help doctors better understand and trust the AI-generated results during diagnosis.

Chapter 6

Timeline

This chapter presents the project timeline and major milestones.



Figure 6.1: Project timeline and milestones.

Chapter 7

Discussion

Our project investigated an interpretable sparse dictionary learning framework for abnormality classification and localization in 3D chest CT scans. A key contribution is the frozen dictionary approach, where normal anatomical patterns are learned first and abnormalities are learned separately. This helps isolate pathological features and improves interpretability.

Unlike many stateoftheart deep learning models that rely on posthoc explainability methods such as GradCAM, the proposed LC-KSVD framework provides inherent explainability. The generated predictions can be directly linked to dictionary atoms and reconstructed voxelspace regions through atom backprojection, allowing more transparent clinical interpretation.

The experimental observations show that the framework can successfully classify and localize pulmonary nodules, Ground Glass Opacity (GGO), and consolidation patterns from 3D chest CT scans while maintaining competitive performance. The proposed objectives of implementing an interpretable dictionary learning pipeline, generating localization maps, and evaluating the framework against existing approaches were successfully achieved.

In realworld clinical applications, this approach could support radiologists by providing both diagnostic predictions and anatomically grounded explanations, improving trust and transparency in AIassisted diagnosis systems.

However, the proposed method has several limitations. The framework is currently limited to three abnormalities and does not yet provide a fully unified endtoend classification and segmentation architecture. In addition, although the explainability is stronger, the classification accuracy may still be lower than largescale deep learning foundation models. Future work will focus on improving performance, expanding abnormality coverage, and integrating more advanced hybrid architectures.

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